

UNIT-5

Stochastic Processes and Markov Chains

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- Syllabus:

Introduction to Stochastic Processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n – step transition probabilities, Markov Chain, Steady state condition, Markov analysis.

- Stochastic Process:

Families of random variables which are functions of say time t , are known as Stochastic processes or random processes or random functions.

Mathematically, a stochastic process is a set of all random variables $\{X(t), t \in T\}$ depending on some real parameter like time 't'

The values assumed by random variables $X(t)$ are called 'states' and the set of all possible values forms the state space of the process.

Example 1: Consider a Random event occurring in time, such as number of telephone calls received at a switch board. Suppose $X(t)$ is the random variable which represents the number of incoming calls in an interval $(0,t)$ of duration t units. The number of calls within a fixed interval of time, say one unit of time, is a random variable $X(t)$ and the family $\{X(t), t \in T\}$ constitutes a stochastic process ($T = [0, \infty]$).

Example 2: Consider a sample experiments like throwing a die. Suppose that X_n is the outcome of the n^{th} throw, $n \geq 1$. Then $\{X_n, n \geq 1\}$ is a family of random variable such that for a distinct value of n , one gets a distinct random X_n ; So $\{X_n, n \geq 1\}$ constitutes a stochastic process known as Bernoulli process

Example 3: _____ A patient's heart pulse during surgery.
Measured continuously during interval $[0, T]$.
Stochastic variable X_t represents the occurrence of
a heartbeat at time t , $0 \leq t \leq T$. Hence,
 X_t assumes only the values 0 (no heartbeat) and 1 (heartbeat)

A stochastic process is said to be discrete if its state space is discrete. A discrete state process is also called chain. Also, if the index set t is discrete, then we have a discrete parameter process. Otherwise we have a continuous parameter process. A discrete parameter process is also called stochastic sequence $\{X_n\}, n \in T$.

Classification of stochastic processes:

1. Continuous stochastic process
2. Discrete stochastic process
3. Deterministic (random) stochastic process
4. Non-deterministic stochastic process

Markov Process: If $\{X(t), t \in T\}$ is a stochastic process such that, given the value $X(s)$, the value of $X(t), t > s$, do not depend on the value of $X(u), u < s$, then the process is said to be a Markov process.

If for $t_1 < t_2 < \dots < t_n < t$

$$P\{a \leq X(t) \leq b / X(t_1) = x_1, X(t_2) = x_2, \dots, X(t_n) = x_n\} = P\{a \leq X(t) \leq b / X(t_n) = x_n\}$$

Then the process $\{X(t), t \in T\}$ is a Markov Process

Markov Chain: Let $X(t)$ be a Markov process which takes only discrete values whether 't' is discrete or continuous is called as Markov Chain.

Example: Consider number of heads obtained in the first n tosses of a coin. The possible values are 0,1,2,.....n.

$$\text{Let } S_{n+1} = \begin{cases} x+1 & \text{if } (n+1)^{th} \text{ trial results in head} \\ x & \text{if } (n+1)^{th} \text{ trial results in trial} \end{cases}$$

$$S_n = X_0 + X_1 + X_2 + \cdots + X_n$$

i.e the conditional probability of S_{n+1} given S_n depends on S_n but not on the manner how the value of S_n is derived.

If we let $P[X_n = a_n / X_{n-1} = a_{n-1}, X_{n-2} = a_{n-2}, \dots, X_1 = a_1]$

Such that $P[X_n = a_n / X_{n-1} = a_{n-1}]$ for every n.

A Markov chain is a stochastic Model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event.

i.e Consider a system which can be in any one of a finite number of states $E_1, E_2, \dots, E_n$. We also assume that the probability of the system being in a given state at the next trial depends only on its present state and not upon the states it may have been in earlier times. If any time, the system is in a state E_i , the probability of it being in the state E_j at the next trial is P_{ij} .

$$\text{i.e } P\{(E_i, E_j)\} = a_i P_{ij}$$

$$\text{Similarly } P\{(E_i, E_j, E_k)\} = a_i P_{ij} P_{jk}$$

Where a_i is the probability distribution of E_i

Here P_{ij} is called the **probability of a transition** from E_i

The transition probabilities P_{ij} will be arranged in a matrix form is called **matrix of transition probabilities** or **Transition Probability matrix**

Stochastic Matrix:

The transition probabilities P_{ij} will be arranged in a matrix of transition probabilities

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} & \cdots & \cdots \\ P_{21} & P_{22} & P_{23} & \cdots & \cdots \\ P_{31} & P_{32} & P_{33} & \cdots & \cdots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix}$$

Clearly P is a square matrix with non-negative elements and unit row sums. Such a matrix (finite or infinite) is called a **Stochastic matrix**

NOTE:

1. A stochastic matrix P is said to be **regular** if all the entries of some power P^m are positive.
2. A stochastic matrix P is not regular if '1' occurs in the principal main diagonal.
3. If P and Q are stochastic matrices then product PQ is also a stochastic matrix.

Problems:

I. Which of the following matrices are stochastic

$$(i) \quad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$(ii) \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(iii) \quad C = \begin{bmatrix} 0 & 1 \\ 1/3 & 1/4 \end{bmatrix}$$

$$(iv) \quad D = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$(v) \quad E = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$$

Soln:

(i) A is not a square matrix

$\therefore$ It is not stochastic

(ii) The matrix is a square matrix with non-negative entries and sum of the elements in each row is equal to 1

$\therefore$ B is stochastic

(iii) C is a square matrix but sum in each row is not equal to 1. So it is not stochastic

(iv) D is stochastic

(v) E is not stochastic , because it contains negative elements.

I. Which of the following matrices are regular.

$$(i) \quad A = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$(ii) \quad B = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$$

$$(iii) \quad C = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{bmatrix}$$

Solution:

(i) A is not a regular matrix since 1 lies on the main diagonal.

$$(ii) \quad B^2 = B \cdot B = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 3/8 & 3/8 & 1/4 \end{bmatrix}$$

$$B^3 = B^2 \cdot B = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 7/16 & 7/16 & 1/8 \end{bmatrix}$$

Since some entries are zero, B is not Regular.

$$(i) C^2 = C \cdot C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

$$C^3 = C^2 \cdot C = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

$$C^4 = C^3 \cdot C = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \\ 1/4 & 1/2 & 1/4 \end{bmatrix}$$

$$C^5 = C^4 \cdot C = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/8 & 1/2 & 3/8 \\ 1/4 & 1/4 & 1/2 \end{bmatrix}$$

Since all the entries of some power of C are positive, C is regular stochastic matrix.

First order and higher order Markov process:

The **Markov chain** of the **first order** is one for which each subsequent state depends only on the immediately preceding one. **Markov** chains of **second** or **higher orders** are the **processes** in which the next state depends on two or more preceding ones.

n – step transition probabilities:

The state transition probability matrix of a Markov chain gives the probabilities of transitioning from one state to another in a single time unit. Also, define an n - step transition probability matrix $P^{(n)}$ whose elements are the n -step transition probabilities.

A **state vector** is a vector (list) that records the probabilities that the system is in any given state at a particular step of the process.

For example, if we know for sure that it is raining today, then the state vector for today will be

$(1, 0)$. But tomorrow is another day! We only know there's a 40% chance of rain and 60% chance of no rain, so tomorrow's state vector is $(0.4, 0.6)$.

Classification of states:

A **state** is any particular situation that is possible in the system. For example, if we are studying rainy days, then there are two states:

1. It's raining today.
2. It's not raining today.

The system could have many more than two states also.

There are 3 states in Markov process

Recurrent State: A state i is said to be recurrent if

$$\sum P_{ii}^{(n)} = 1$$

Transient State: If there exist j and n such that

$\sum P_{ij}^{(n)} > 0$ but $\sum P_{ji}^{(m)} = 0$ for all m , then state 'i' is called transient state. i.e it process the property that it is possible with positive probability to pass from it to another state but it is not possible to return from that state to original state.

Irreducible State: If the two states i & j are such that, each is accessible from the other, then we say that the two states communicate. It is denoted by $i \leftrightarrow j$. Then $\exists$ an integer m & n such that

$$\sum P_{ij}^{(n)} > 0 \quad \text{and} \quad \sum P_{ji}^{(m)} > 0$$

Problems:

Three Boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probability that

- (i) A has the ball
- (ii) B has the ball
- (iii) C has the ball after 3 throw

Solution:

$P_{11} \rightarrow$ Probability that the ball is thrown from A to A = 0

$P_{12} \rightarrow$ Probability that the ball is thrown from A to B = 1

$$P_{13} = 0, P_{21} = 0, P_{22} = 0, P_{23} = 1, P_{31} = \frac{1}{2}, P_{32} = \frac{1}{2}, P_{33} = 0$$

Then Transition Matrix $P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$

It is given 3 throws, so calculate P^3

$$P^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix}$$

$$\begin{aligned} P^3 &= P^2 \cdot P = \begin{bmatrix} 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} \end{aligned}$$

Therefore the ball was first thrown by C, the initial probability distribution is given by row matrix

$$A^{(0)} = [P_1^{(0)} \quad P_2^{(0)} \quad P_3^{(0)}] = [0 \quad 0 \quad 1]$$

After 3 throws, the probability distribution is given by

$$\begin{aligned} [P_1^{(3)} \quad P_2^{(3)} \quad P_3^{(3)}] &= A^{(0)} P^3 = [0 \quad 0 \quad 1] \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/4 & 1/4 & 1/2 \end{bmatrix} \\ &= \left[\frac{1}{4} \quad \frac{1}{4} \quad \frac{1}{2} \right] \end{aligned}$$

After 3 throws, the probability that the ball is with

- (i) $A = \frac{1}{4}$
- (ii) $B = \frac{1}{4}$
- (iii) $C = \frac{1}{2}$

2. A person owning a scooter has the option to switch over to scooter, bike or a car next time with the probability of $[0.3 \quad 0.5 \quad 0.2]$. If the transition probability matrix is $\begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$ what are the probability of vehicles elated to his fourth purchase.

Solution:

$$\text{Given } P = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

We have to calculate for third purchase

$$P^2 = P.P = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.295 & 0.345 & 0.36 \\ 0.255 & 0.385 & 0.36 \\ 0.275 & 0.325 & 0.4 \end{bmatrix}$$

$$P^3 = P^2.P = \begin{bmatrix} 0.295 & 0.345 & 0.36 \\ 0.255 & 0.385 & 0.36 \\ 0.275 & 0.325 & 0.4 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.2 & 0.5 & 0.3 \\ 0.25 & 0.25 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.277 & 0.351 & 0.372 \\ 0.269 & 0.359 & 0.372 \\ 0.275 & 0.345 & 0.380 \end{bmatrix}$$

Therefore probability of his fourth purchase is $[0.3 \quad 0.5 \quad 0.2]P^3$

$$= [0.3 \quad 0.5 \quad 0.2] \begin{bmatrix} 0.277 & 0.351 & 0.372 \\ 0.269 & 0.359 & 0.372 \\ 0.275 & 0.345 & 0.380 \end{bmatrix}$$

$$= [0.2726 \quad 0.3538 \quad 0.3736]$$

1. Consider a Markov Process with state space $S = \{0,1,2\}$ and

$$\text{transition matrix } P = \begin{bmatrix} p & q & 0 \\ 1/2 & 0 & 1/2 \\ p - 1/2 & 7/10 & 1/5 \end{bmatrix}$$

- (i) What can you say about the values of p & q ?
- (ii) Calculate the transition probability $P_{ij}^{(3)}$

Solution:

- (i) From the third row if we sum, all the elements it should be equal to 1

$$\text{i.e. } p - \frac{1}{2} + \frac{7}{10} + \frac{1}{5} = 1 \Rightarrow p = \frac{3}{5}$$

$$\text{put this in the first row} \Rightarrow p + q = 1 \Rightarrow q = \frac{2}{5}$$

(ii) Transition probability

$$\begin{aligned} &= P_{ij}^{(3)} \\ &= \begin{bmatrix} 3/5 & 2/5 & 0 \\ 1/2 & 0 & 1/2 \\ 1/10 & 7/10 & 1/5 \end{bmatrix} \begin{bmatrix} 3/5 & 2/5 & 0 \\ 1/2 & 0 & 1/2 \\ 1/10 & 7/10 & 1/5 \end{bmatrix} \begin{bmatrix} 3/5 & 2/5 & 0 \\ 1/2 & 0 & 1/2 \\ 1/10 & 7/10 & 1/5 \end{bmatrix} \\ &= \begin{bmatrix} 0.476 & 0.364 & 0.160 \\ 0.495 & 0.210 & 0.295 \\ 0.387 & 0.445 & 0.168 \end{bmatrix} \end{aligned}$$

Classification of Chains:

Here is the following classifications

Absorbing Chain: A Markov chain is absorbing if

- (i) It has at least one absorbing state. Here absorbing state means suppose j is absorbing iff $P_{jj} = 1$ & $P_{jk} = 0, k \neq j$
- (ii) It is possible to go from every non absorbing state to at least one absorbing state.

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix} \text{ Here first and last states are absorbing}$$

Ergodic Chain: a chain in which every two states can communicate with each other and in which the system cannot be left is called ergodic chain.

A special chain of an ergodic chain is the regular chain in which for some n , P^n has no zero elements

Irreducible Chain: A Markov chain in which every state can be reached from every other state. Otherwise reducible.

All the states are reachable to one another so the matrix is Ergodic

1. The transition probability matrix of a Markov chain is given by

$$p = \begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix}. \text{ Is this matrix irreducible?}$$

Solution:

Consider the three states as 0,1,2.

In this chain we go from state 0 to state 1 with a probability of 0.7 and from state 1 to state 2 with a probability 0.5

Thus it is possible to go from state 0 to state 2

$\therefore$ The chain is irreducible and all the states are recurrent.

Theorem :- *

A Stochastic matrix P is not regular if 1 occurs in the principle main diagonal.

1. Which of the following matrices are stochastic.

(i) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (ii) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$ (v) $\begin{bmatrix} 0 & 2 \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}$

Sol (i) The matrix is a square matrix with non-negative entries and sum of elements in each row is equal to 1

$\therefore \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ & $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ are stochastic matrices

(iii) It is not a square matrix

$\therefore$ It is not stochastic.

(iv) The matrix is not stochastic, because it contains negative elements

(v) The matrix is square matrix but sum in each row is not equal to 1

$\therefore$ It is not a stochastic matrix.

2) Which of the Stochastic matrices are regular.

$$(i) A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$(ii) \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

$$(iii) \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$(iv) \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 1 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol (i) $G.T \quad A = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$

Not Regular since 1 lies on the main diagonal

$$(ii) G.T \quad B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

$$B^2 = B \cdot B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{8} & \frac{3}{8} & 0 \end{bmatrix}$$

$$B^3 = B^2 \cdot B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{7}{16} & \frac{7}{16} & \frac{1}{8} \end{bmatrix}$$

Since entries b_{13}, b_{23} are zero

$\therefore B$ is not regular.

$$(iii) G.T. \quad C = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\therefore C^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}, \quad C^3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$C^4 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}, \quad C^5 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$

Since all entries of some powers of C are positive

$\therefore C$ is regular stochastic process.

$$(iv) G.T. \quad D = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & 1 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Not Regular since 1 lies on the main diagonal.

3) Represent the following transition matrices as a diagram

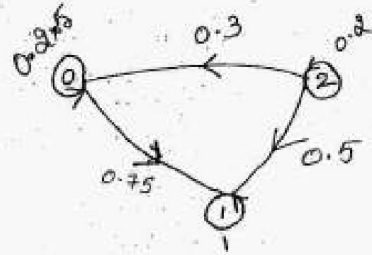
$$(i) \begin{bmatrix} 0.25 & 0.75 & 0 \\ 0 & 1 & 0 \\ 0.3 & 0.5 & 0.2 \end{bmatrix}$$

$$(ii) \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

Sol:- (i) G.T the TPM of Markov chain is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.25 & 0.75 & 0 \\ 0 & 1 & 0 \\ 0.3 & 0.5 & 0.2 \end{bmatrix} \end{matrix}$$

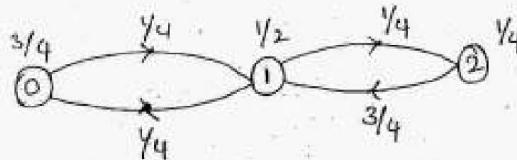
The diagram of chain is



(ii) G.T, TPM of Markov chain is

$$P = \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix}$$

$\therefore$ The diagram of chain is



4) Suppose that the probability of a dry day (state 0) following a rainy day (state 1) is $\frac{1}{3}$ and that the probability of a rainy day following a dry day is $\frac{1}{2}$. Given that May 1st is a dry day. Find the probability (i) May 3rd is also a dry day & (ii) May 5 is also a dry day.

dry day = 0
rainy day = 1

Sol:- we have two-state Markov chain such that

$P_{10} = \frac{1}{3}$ & $P_{01} = \frac{1}{2}$ & t.p.m is given by

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix} \end{matrix}$$

$$P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

we have $P^2 = \begin{bmatrix} \frac{5}{12} & \frac{7}{12} \\ \frac{7}{18} & \frac{11}{18} \end{bmatrix}$

$$P^4 = \begin{bmatrix} \frac{173}{432} & \frac{259}{432} \\ \frac{259}{648} & \frac{389}{648} \end{bmatrix}$$

Given that, may 1 is a dry day, then the probability that may 3 is a dry day is $\frac{5}{12}$ & that may 5 is a dry day is $\frac{173}{432}$

5) A training process is considered as a two-state markov chain. If it rains, it is considered to be in state 0 & it does not rain, The chain is in the state of 1. The transition probability of the markov chain is defined by $P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$. Find the probability that it will rain for 3 days from today assuming that it is raining today. Assume that the mutual probabilities of state 0 (or) state 1 as 0.4 & 0.6 respectively.

Sol:- G.T. the one step t.p.m is given by

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$$

$$P^2 = P \cdot P$$

$$= \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 0.44 & 0.56 \\ 0.28 & 0.72 \end{bmatrix}$$

$$P^3 = P^2 \cdot P = \begin{bmatrix} 0.44 & 0.56 \\ 0.28 & 0.72 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.8 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.376 & 0.624 \\ 0.312 & 0.688 \end{bmatrix}$$

The probability that it will rain on 3rd day, given that it will rain today is 0.376.

6) Let $\{x_n, n \geq 0\}$ be a Markov chain with

Three states $\{0, 1, 2\}$ & with transition matrix

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix} \end{matrix}$$

& the initial distribution: $P\{x_0 = i\} = \frac{1}{3}; i = 0, 1, 2$. Find $P\{x_3 = 1, x_2 = 2, x_1 = 1, x_0 = 2\}$

Sol:- G.T, the tpm of the Markov Chain is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{4} \end{bmatrix} \end{matrix} \text{ \& } P\{x_0 = i\} = \frac{1}{3}, i = 0, 1, 2$$

Then we have

$$P\{x_1 = 1 / x_0 = 2\} = \frac{1}{4} = p_{21}$$

$$P\{x_2 = 2 / x_1 = 1\} = \frac{1}{4} = p_{12}$$

$$P\{x_2=2, x_1=1/x_0=2\} = P\{x_2=2/x_1=1\} \cdot P\{x_1=1/x_0=2\}$$

$$= \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$$

$$\therefore P\{x_2=2, x_1=1, x_0=2\} = P\{x_2=2, x_1=1/x_0=2\} \cdot P\{x_0=2\}$$

$$= \frac{3}{16} \cdot \frac{1}{8} = \frac{1}{16}$$

$$\therefore P\{x_3=1, x_2=2, x_1=1, x_0=2\} = P\{x_3=1/x_2=2, x_1=1, x_0=2\}$$

$$\cdot P\{x_2=2, x_1=1, x_0=2\}$$

$$= P\{x_3=1/x_2=2\} \cdot P\{x_2=2, x_1=1, x_0=2\}$$

$$= \frac{3}{4} \cdot \frac{1}{16} = \frac{3}{64}$$

7) The t.p.m of a markov chain $\{x_n\}$, $n=1,2,3,\dots$ having 3 states 1, 2 & 3 is $P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ and the initial distribution is $p^{(0)} = (0.7, 0.2, 0.1)$ Find (i) $P\{x_2=3\}$ (ii) $P\{x_3=2, x_2=3, x_1=3, x_0=2\}$.

Sol G.T the t.p.m of markov chain is

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \end{matrix}$$

we have $p(x_0=1)=0.7$, $p(x_0=2)=0.2$, $p(x_0=3)=0.1$

$$\text{Also } p^{(2)} = P^2 = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$$

$$P^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.43 & 0.31 & 0.26 \\ 0.24 & 0.42 & 0.34 \\ 0.36 & 0.35 & 0.29 \end{bmatrix} \end{matrix}$$

$$\begin{aligned} \text{(i)} \quad P(X_2=3) &= \sum_{i=1}^3 P\{X_2=3/X_0=i\} \cdot P\{X_0=i\} \\ &= P\{X_2=3/X_0=1\} \cdot P\{X_0=1\} + P\{X_2=3/X_0=2\} \\ &\quad P\{X_0=2\} + P\{X_2=3/X_0=3\} \cdot P\{X_0=3\} \\ &= P_{13}^{(2)} \cdot P\{X_0=1\} + P_{23}^{(2)} P\{X_0=2\} + P_{33}^{(2)} \cdot P\{X_0=3\} \\ &= 0.26 (0.7) + 0.34 (0.2) + 0.29 (0.1) \\ &= 0.279. \end{aligned}$$

$$\text{(ii)} \quad P\{X_1=3/X_0=2\} = P_{23} = 0.2$$

$$P\{X_1=3, X_0=2\} = P\{X_1=3/X_0=2\} \cdot P\{X_0=2\}$$

$$= 0.2 \times 0.2$$

$$= 0.04$$

$$P\{X_2=3, X_1=3, X_0=2\} = P\{X_2=3/X_1=3, X_0=2\} \cdot P\{X_1=3, X_0=2\}$$

$$= P\{X_2=3/X_1=3\} \cdot P\{X_1=3, X_0=2\}$$

$$= 0.3 \times 0.04$$

$$= 0.012 \quad (\text{by markov property})$$

$$P\{x_3=2, x_2=3, x_1=3, x_0=2\}$$

$$= P\{x_3=2/x_2=3, x_1=3, x_0=2\} \cdot P\{x_2=3, x_1=3, x_0=2\}$$

$$= P\{x_3=2/x_2=3\} \cdot P\{x_2=3, x_1=3, x_0=2\}$$

$$= 0.4 \times 0.012$$

$$= 0.0048$$

8) A fair die tossed repeatedly. If x_n denotes the maximum of the numbers occurring in the first n tosses. Find the transition probability matrix P of the Markov chain $\{x_n\}$. Find also P^2 and $P(x_2=6)$.

Sol:- State Space = $\{1, 2, 3, 4, 5, 6\}$

The TPM is formed using the following analysis

Let x_n = The maximum of the numbers occurring in the first

n trials = 3 (say)

Then $x_{n+1} = 3$ if the $(n+1)^{th}$ trial results is 1, 2 or 3.

= 4 if the $(n+1)^{th}$ trial results is 4

= 5 if the $(n+1)^{th}$ trial results is 5

= 6 if the $(n+1)^{th}$ trial results is 6

$$\therefore P\{x_{n+1}=3/x_n=3\} = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{3}{6}$$

$$\therefore P\{x_{n+1}=i/x_n=3\} = \frac{1}{6} \text{ when } i = 4, 5, 6$$

i. The TPM of chain is x_{n+1}

$$P = \begin{bmatrix} \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & \frac{2}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & \frac{5}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & \frac{4}{6} & \frac{1}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & \frac{5}{6} & \frac{1}{6} \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\frac{5}{6} = 1$$

for
the
prob
of
pr

$$(ii) P^2 = \frac{1}{36} \begin{bmatrix} 1 & 3 & 5 & 7 & 9 & 11 \\ 0 & 4 & 5 & 7 & 9 & 11 \\ 0 & 0 & 9 & 7 & 9 & 11 \\ 0 & 0 & 0 & 16 & 9 & 11 \\ 0 & 0 & 0 & 0 & 25 & 11 \\ 0 & 0 & 0 & 0 & 0 & 36 \end{bmatrix}$$

Initial state probability distribution is $P^{(0)} = (\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$

(Since all the values of 1, 2, ..., 6 are equally likely)

$$(ii) P(X_2=6) = \sum_{i=1}^6 P(X_2=6/X_0=i) P(X_0=i)$$

$$= \frac{1}{6} \sum_{i=1}^6 P_{i6}^{(2)}$$

$$= \frac{1}{6} [P_{16}^{(2)} + P_{26}^{(2)} + P_{36}^{(2)} + P_{46}^{(2)} + P_{56}^{(2)} + P_{66}^{(2)}]$$

$$= \frac{1}{6} \left[\frac{11}{36} + \frac{11}{36} + \frac{11}{36} + \frac{11}{36} + \frac{11}{36} + \frac{11}{36} \right]$$

$$= \frac{1}{6 \times 36} [11 + 11 + 11 + 11 + 11 + 11] = 9)$$

$$P(X_2=6) = \frac{91}{216}$$

9) A gambler has Rs. 2. He bets Rs. 1 at a time and wins Rs. 1 with probability $\frac{1}{2}$. He stops playing if he loses Rs. 2.

(or) wins Rs. 4.

(a) What is the TPM of related Markov chain?

(b) What is the probability that he has lost his money at the end of 5 plays?

(c) What is the probability that the game lasts more than 7 plays?

Sol Let X_n represent the amount with the player at the end of the n th round of the play.

The state space $X_n = \{0, 1, 2, 3, 4, 5, 6\}$

When the game is stopped, if the player loses Rs. 2, $X_n = 0$

(or) If he wins Rs. 4, $X_n = 6$.

The TPM of Markov chain is written as

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

This markov chain is called random walks with absorbing barriers at 0 & 6, since the chain cannot come out of the states 0 & 6, once it has entered.

b) Since the player has Rs. 2 to start the play, the initial probability distribution of x_n is

$$p^{(0)} = (0 \ 0 \ 1 \ 0 \ 0 \ 0)$$

probability distribution after one play is given by

$$p^{(1)} = p^{(0)} \cdot p = (0 \ 0 \ 1 \ 0 \ 0 \ 0) p = (0 \ \frac{1}{2} \ 0 \ \frac{1}{2} \ 0 \ 0)$$

$$p^{(2)} = p^{(1)} \cdot p = (0 \ \frac{1}{2} \ 0 \ \frac{1}{2} \ 0 \ 0) p = (\frac{1}{4} \ 0 \ \frac{1}{2} \ 0 \ \frac{1}{4} \ 0)$$

$$p^{(3)} = p^{(2)} \cdot p = (\frac{1}{4} \ 0 \ \frac{1}{2} \ 0 \ \frac{1}{4} \ 0) p = (\frac{1}{4} \ \frac{1}{4} \ 0 \ \frac{3}{8} \ 0 \ \frac{1}{8})$$

$$p^{(4)} = p^{(3)} \cdot p = (\frac{1}{4} \ \frac{1}{4} \ 0 \ \frac{3}{8} \ 0 \ \frac{1}{8}) p = (\frac{3}{8} \ 0 \ \frac{5}{16} \ 0 \ \frac{1}{4} \ \frac{1}{16})$$

$$p^{(5)} = p^{(4)} \cdot p = (\frac{3}{8} \ 0 \ \frac{5}{16} \ 0 \ \frac{1}{4} \ \frac{1}{16}) p = (\frac{3}{8} \ \frac{5}{32} \ 0 \ \frac{9}{32} \ 0 \ \frac{1}{8})$$

The probability that the player has lost his money at the end of 5 plays $= p(x_5 = 0) = \frac{3}{8}$

c) The probability that the game lasts more than 7 plays

$= p(\text{system is neither in state 0 nor in 6 at the end of the seventh round})$.

$$\text{using } p^{(6)} = p^{(5)} \cdot p = \begin{pmatrix} \frac{29}{64} & 0 & \frac{7}{32} & 0 & \frac{13}{64} & 0 & \frac{1}{8} \end{pmatrix}$$

$$p^{(7)} = p^{(6)} \cdot p = \begin{pmatrix} \frac{29}{64} & \frac{1}{64} & \frac{23}{64} & 0 & \frac{13}{128} & \frac{1}{128} \end{pmatrix}$$

$$\therefore p\{x_7 = 1, 2, 3, 4 \text{ or } 5\} = \frac{7}{64} + 0 + \frac{27}{128} + 0 + \frac{13}{128}$$

$$= \frac{27}{64}$$

10) Three boys A, B & C are Throwing a ball to each other. A always throws the ball to B & B always throws the ball to C; but C is just as likely to throw the ball to B as A. Show that the process is markovian. Find the transition matrix & classify the states. Are all the states ergodic?

Sol:- The transition probability matrix of the process $\{X_n\}$ is given below.

$$P_{(present)} = \begin{matrix} & \begin{matrix} \text{States of } X_n \text{ (future)} \\ \begin{matrix} A & B & C \end{matrix} \end{matrix} \\ \begin{matrix} \text{States of } X_{n-1} \\ \begin{matrix} A \\ B \\ C \end{matrix} \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix} \end{matrix}$$

State of X_n depends only on state of X_{n-1} but not on states of $X_{n-2}, X_{n-3}, \dots$ or earlier states.

$\therefore \{X_n\}$ is a markov chain.

Now $P^2 = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, $P^3 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$

$P_{11}^{(3)} > 0$, $P_{13}^{(2)} > 0$, $P_{21}^{(2)} > 0$, $P_{22}^{(2)} > 0$, $P_{33}^{(2)} > 0$ and all other

$\therefore$ The chain is Irreducible.

Irreducible :-

$P^4 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{bmatrix}$, $P^5 = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{8} & \frac{1}{2} \end{bmatrix}$ and so on
 $P_{ij}^{(n)} > 0$.

$P^6 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{8} & \frac{1}{2} \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} \end{bmatrix}$ & so on.

We note that $P_{11}^{(2)}$, $P_{11}^{(3)}$, $P_{11}^{(4)}$, $P_{11}^{(5)}$, $P_{11}^{(6)}$ etc are > 0 . for $i=2,3$

G.C.D of 2, 3, 4, 5, 6, ... = 1

The states 2 & 3 (i.e. B & C) are periodic with period 1
i.e. aperiodic.

We note that $P_{11}^{(3)}$, $P_{11}^{(5)}$, $P_{11}^{(6)}$ etc, are > 0 . & G.C.D of
3, 5, 6, ... = 1

$\therefore$ The state 1 (i.e. state A) is periodic with period 1 i.e.
aperiodic.

Since the chain is finite and irreducible, all st

States are non-null persistent.

Hence, all the states are ergodic.

11) Find the nature of states of the Markov chain with TPM

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Sol:- G.T the time of Markov chain is

$$P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$P^2 = P \cdot P = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$P^3 = P^2 \cdot P = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix} = P$$

$$P^4 = P^3 = P = P \cdot P = P^2$$

$$\text{// } P^5 = P^3 \cdot P^2 = P \cdot P^2 = P$$

$$P^6 = P^4 \cdot P^2 = P^2 \cdot P^2 = P^2$$

$$\text{In general } P^{2n} = P^2 \text{ \& } P^{2n+1} = P \cdot P = P^2 \cdot P = P^3 = P$$

Also, we observe

$$P_{00}^{(2)} > 0, P_{02}^{(2)} > 0, P_{11}^{(2)} > 0, P_{20}^{(2)} > 0, P_{22}^{(2)} > 0.$$

$$P_{01}^{(1)} > 0, P_{10}^{(1)} > 0, P_{12}^{(1)} > 0, P_{21}^{(1)} > 0.$$

Hence the markov chain is irreducible.

Also $P_{ii}^{(2)} > 0, P_{ii}^{(4)} > 0, P_{ii}^{(6)} > 0$ & so on for every 'i'.

Thus $P_{ii}^{(2n)} > 0, P_{ii}^{(2n+1)} = 0$, for each 'i'.

$\therefore$ All the states of the markov chain are periodic with period 2.

Since the markov chain is finite & irreducible, all its states are non-null persistent. All the states are not ergodic.

* Irreducible:- If $P_{ij}^{(n)} > 0$ for some n & $\forall i \neq j$. Then every state can be reached from every other state. When this condition is satisfied. The markov chain is said to be irreducible. The TPM of an irreducible chain is an irreducible-matrix otherwise the chain is said to be reducible.

19) The transition probability matrix of a markov chain is given

by $\begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix}$. Is this matrix irreducible.

Sol Consider the Three states as 0, 1, 2

$$\begin{matrix} & \begin{matrix} 0 & 1 & 2 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \end{matrix} & \begin{bmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.4 & 0.5 \\ 0 & 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

In this chain we go from state 0 to state 1 with probability of 0.7 & from state 1 to state 2 with probability 0.5. Thus it is possible to go from state 0 to state 2.

$\therefore$ The chain is Irreducible & all the states are recurrent.

3. If the matrix $\begin{bmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ is irreducible.

Sol $P = \begin{bmatrix} 0.4 & 0.6 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 \\ 0.2 & 0.4 & 0.1 & 0.3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$P^2 = P \cdot P = \begin{bmatrix} 0.34 & 0.66 & 0 & 0 \\ 0.33 & 0.67 & 0 & 0 \\ 0.22 & 0.44 & 0.01 & 0.33 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.334 & 0.666 & 0 & 0 \\ 0.333 & 0.667 & 0 & 0 \\ 0.222 & 0.444 & 0.001 & 0.333 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

W.K.T, one of the rule to prove that, a matrix is Irreducible is that the sum of each row of the matrix must be equal to 1, but in matrix P^3 , the sum of third row is not equal to 1. Hence The given matrix is not Irreducible.

* If $x = [x_1, x_2, \dots, x_n]$ is the steady-state distribution of the chain whose tpm is n^{th} order square matrix 'p'.

$$\text{Then } x = xp.$$

14) If the tpm of a marker chain is $\begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$, find the steady state distribution of the chain.

sol let $x = [x_1, x_2]$ is the steady-state distribution of the chain

$$\text{Then } x = xp \text{ and } x_1 + x_2 = 1 \text{ --- (1)}$$

$$[x_1 \ x_2] = [x_1 \ x_2] \begin{bmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$[x_1 \ x_2] = [\frac{1}{2}x_2 \quad x_1 + \frac{1}{2}x_2]$$

$$x_1 = \frac{1}{2}x_2 \Rightarrow 2x_1 = x_2 \text{ --- (2)}$$

$$x_2 = x_1 + \frac{1}{2}x_2 \text{ --- (3)}$$

sub eqⁿ (2) in (1), we get

$$x_1 + 2x_1 = 1$$

$$3x_1 = 1$$

$$x_1 = \frac{1}{3}$$

from eqⁿ (2), we get

$$x_2 = 2 \times \frac{1}{3}$$

$$= \frac{2}{3}$$

$\therefore$ Hence $\left[\frac{1}{3} \quad \frac{2}{3} \right]$ be the steady state distribution of the chain.

15) Study of passage of english text to find a vowel followed by a vowel or a consonant followed by a consonant or a vowel reveal the following transition probability matrix $\begin{bmatrix} 0.12 & 0.88 \\ 0.54 & 0.46 \end{bmatrix}$. Find the percentage of letters in the text book which are expected to be vowels.

Sol Given that the transition probability matrix $P = \begin{bmatrix} 0.12 & 0.88 \\ 0.54 & 0.46 \end{bmatrix}$

we have to find the limiting probabilities

$$X = XP \text{ \& } x_1 + x_2 = 1 \quad \text{--- (1)}$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 0.12 & 0.88 \\ 0.54 & 0.46 \end{bmatrix}$$

$$x_1 = 0.12x_1 + 0.54x_2 \quad \text{--- (2)}$$

$$x_2 = 0.88x_1 + 0.46x_2 \quad \text{--- (3)}$$

$$\text{Sub, } x_2 = 1 - x_1 \text{ in (2) (from (1))}$$

we get

$$0.12x_1 + 0.54(1 - x_1) = x_1$$

$$\Rightarrow 0.12x_1 - 0.54x_1 + 0.54 = x_1$$

$$\Rightarrow x_1 = 0.3803$$

$$\therefore x_2 = 1 - x_1$$

$$= 1 - 0.3803$$

$$x_2 = 0.6197$$

$\therefore$ Hence, the percentage of letters in the textbook which are expected to be vowels is 38%.

16) The school of International Studies for population found out by its survey that the mobility of the population of a state to village, town and city is the following percentage.

From \ To	village	Town	city
village	30%	20%	50%
Town	30%	50%	20%
city	10%	40%	50%

What will the proportion of population in village, town & city after two years? present population has proportion of 0.4, 0.3 and 0.3 village, town and city respectively. Find the proportions in the long run.

Sol:- Given that the school of International studies for population found to its survey that the mobility of the population of a state to village, Town & City are.

From \ To	village	Town	city
village	30%	20%	50%
Town	30%	50%	20%
city	10%	40%	50%

which can be written as

$$P = \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

The present population proportion $P_0 = [0.4, 0.3, 0.3]$

The proportion of population in village, town & city after one year is

$$\begin{aligned} P_1 &= P_0 \cdot P \\ &= [0.4, 0.3, 0.3] \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix} \end{aligned}$$

$$P_1 = \begin{bmatrix} 0.24 & 0.35 & 0.4 \end{bmatrix}$$

Hence the proportion of population in village, town and city after one year

$$P_1 = \begin{bmatrix} 0.24 & 0.35 & 0.4 \end{bmatrix}$$

→ The proportion of population in village, town and city after two years

$$P_2 = P_1 \cdot P$$

$$= \begin{bmatrix} 0.24 & 0.35 & 0.4 \end{bmatrix} \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0.218 & 0.387 & 0.395 \end{bmatrix}$$

Hence the proportion of population in village, town and city after two years are 0.218, 0.387 & 0.395.

(b) The proportion of population in village, town and city in long run.

i.e the steady-state distribution of the chain, then

$$X = X P \text{ and } x_1 + x_2 + x_3 = 1$$

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.3 & 0.5 & 0.2 \\ 0.1 & 0.4 & 0.5 \end{bmatrix}$$

$$x_1 = 0.3x_1 + 0.3x_2 + 0.1x_3$$

$$\Rightarrow 0.7x_1 = 0.3x_2 + 0.1x_3 \text{ --- (1)}$$

$$x_2 = 0.2x_1 + 0.5x_2 + 0.4x_3$$

$$\Rightarrow 0.5x_2 = 0.2x_1 + 0.4x_3 \text{ --- (2)}$$

$$x_3 = 0.5x_1 + 0.2x_2 + 0.5x_3$$

$$\Rightarrow 0.5x_3 = 0.5x_1 + 0.2x_2 \text{ --- (3)}$$

$$x_1 + x_2 + x_3 = 1$$

$$x_1 = 1 - x_2 - x_3 \text{ --- (4)}$$

Now multiplying eqⁿ (4) with 0.7 and subtracting eqⁿ (1) we get,

$$0.7x_1 = 0.7 - 0.7x_2 - 0.7x_3$$

$$0.7x_1 = 0 + 0.3x_2 + 0.1x_3$$

$$0 = 0.7 - 1.0x_2 - 0.8x_3$$

$$\therefore x_2 + 0.8x_3 = 0.7 \text{ --- (5)}$$

Now multiply eqⁿ (2) with 0.5 and eqⁿ (3) with 0.2 then we get:

$$0.25x_2 = 0.1x_1 + 0.2x_3$$

$$0.1x_3 = 0.1x_1 + 0.04x_2$$

$$0.25x_2 - 0.1x_3 = 0.2x_3 - 0.04x_2$$

$$0.25x_2 + 0.04x_2 = 0.2x_3 + 0.1x_3$$

$$0.29x_2 = 0.3x_3$$

$$x_2 = \left(\frac{0.3}{0.29}\right)x_3$$

$$x_2 = (1.034)x_3 \text{ --- (6)}$$

Now, substituting eqⁿ (6) in eqⁿ (5), we get

$$(1.034)x_3 + 0.8x_3 = 0.7$$

$$1.834x_3 = 0.7$$

$$x_3 = \frac{0.7}{1.834}$$

$$x_3 = 0.382$$

Sub $x_3 = 0.382$ in eqⁿ (6), we get

$$x_2 = (1.034)(0.382)$$

$$x_2 = 0.394$$

put $x_2 = 0.394$ and $x_3 = 0.382$ in eqⁿ (4), we get

$$x_1 = 1 - 0.394 - 0.382$$

$$x_1 = 0.224$$

$$X = [x_1 \ x_2 \ x_3] = [0.224 \ 0.394 \ 0.382]$$

$\therefore$ The long run proportion of population move from state to village, town and city is 22.4%, 39.4% and 38.2% respectively.

17) If the transition probability matrix of market shares of three brands A, B and C is $\begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$ and the

initial market shares are 50%, 25% and 25%.

Find (i) The market shares in second and third periods.

(ii) The limiting probabilities.

Sol:- Given that, the initial market shares of three brands A, B and C are 50%, 25% and 25% which can be written as $P_0 = [0.5 \quad 0.25 \quad 0.25]$

The transition probability matrix

$$P = \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

The market shares in the second period = $P_0 \cdot P^2$

$$= [0.5 \quad 0.25 \quad 0.25] \begin{bmatrix} 0.5050 & 0.225 & 0.27 \\ 0.435 & 0.275 & 0.29 \\ 0.48 & 0.23 & 0.29 \end{bmatrix}$$

$$= [0.4813 \quad 0.2387 \quad 0.2800]$$

Hence the market shares in the third period = $p_0 \cdot p^3$ (2)

$$= \begin{bmatrix} 0.5 & 0.25 & 0.25 \end{bmatrix} \begin{bmatrix} 0.4145 & 0.2085 & 0.19 \\ 0.4675 & 0.2105 & 0.1985 \\ 0.4652 & 0.2307 & 0.221 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4815 & 0.2382 & 0.2802 \end{bmatrix}$$

Hence the market shares in the third period is

48.1%, 23.8%, 28.02%.

ii) The limiting probability can be interpreted as the long run market shares that can be reached to an equilibrium.

$X = [x_1 \ x_2 \ x_3]$ then $X = Xp$ and $x_1 + x_2 + x_3 = 1$ — (1)

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 0.4 & 0.3 & 0.3 \\ 0.8 & 0.1 & 0.1 \\ 0.35 & 0.25 & 0.4 \end{bmatrix}$$

$$\text{Now } x_1 = 0.4x_1 + 0.8x_2 + 0.35x_3$$

$$0.6x_1 - 0.8x_2 - 0.35x_3 = 0 \text{ — (2)}$$

$$\text{Now } x_2 = 0.3x_1 + 0.1x_2 + 0.25x_3$$

$$-0.3x_1 + 0.9x_2 - 0.25x_3 = 0 \text{ — (3)}$$

$$\text{Now } x_3 = 0.3x_1 + 0.1x_2 + 0.4x_3$$

Solving eq^{ns} ①, ② and ③, we get

$$x = 0.4813, \quad y = 0.2383, \quad z = 0.2804$$

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THANK YOU